Nanotechnology-Enhanced Biosensor for Early Detection of Bacterial Sepsis

December 2025 Monthly Progress Report

Project: Nanotechnology-Enhanced Biosensor for Early Detection of Bacterial Sepsis
Organization: IIT Jodhpur & AIIMS Jodhpur (Joint Program in Medical Technologies)
Industry Partner: TCS
Report Period: December 2025
Submitted by: Dr. Mohamed Zaki Hafeez and Sukesh P

Executive Summary

December 2025 achieved critical procurement milestones and comprehensive literature review. Biomarker MOQs finalized, purchase orders placed, first contingent received and second contingent in transit. Systematic review of optimal SERS configuration for experimental phase. The procurement activities represent a major step toward experimental validation of the SERS-based nano-biosensor platform.

Meetings and Review Sessions:

Collaborative Review Meetings with Supervisory Team

Timeline: Multiple sessions throughout December 2025

Key Discussion Points:

- Strategic planning for next phase of project and translation roadmap.

Outcomes:

- Following comprehensive evaluation of project requirements, analytical standards, the research team finalized minimum order quantities for all three biomarker components
- Molecular simulation and frequency analysis approaches
- Discussions on next-phase experimental strategies
- Identification of critical optimization priorities for project.

Minim order quantity (MOQ) Rationale:

Justification for MOQ Quantities:

- NAM: 0.5-1 mg per experiment (100 mg = 100-200 experiments)
- LPS: 0.1 mg per experiment (10 mg = 100 experiments)
- LTA: 0.05 mg per experiment (5 mg = 100 experiments)

Vendors sell complete vials, so buying packages in mg. The cost efficiency is maintained.

Biomarker	Supplier Form	MOQ	Unit	Est. Cost (INR)
N-Acetylmuramic Acid (NAM)	Analytical standard, lyophilized Purity: ≥98% (HPLC)	100	mg	₹12,000-20,000
Lipopolysaccharide (LPS, E. coli O111:B4)	Phenol extracted, Lyophilized powder Purity: ≥90-95%	10	mg	₹12,000-18,000
Lipoteichoic Acid (LTA, S. aureus, Gram-positive extract)	Purified antigen, solution/lyophilized powder Purity: >95%	10	mg	₹30,000-40,000
Total MOQ Procurement	₹54,000-78,000			
Contingency (15%)	₹12,000-20,000			
Total Recommended Budget	₹66,000-98,000			

Order Placement and Procurement

- Vendor Engagement: Successfully sourced multiple qualified suppliers through quotation process and placed order accordingly.
- Receipt of the first contingent of bacterial biomarker standards and awaiting the second one.

Literature review of optimal SERS configuration for experimental phase:

- Literature review of biomarkers limits, solvents, capture probes and existing studies of NAM based on SERS technique.
- comprehensive review of recent scientific literature to establish the theoretical foundation for experimental work.

December 2025 represents a critical transition point from planning and preparation to active experimental work. With procurement milestones and technical preparation, the research team is looking forward to execute the planned experimental work with confidence and efficiency.